CoFiDA-M: Concept-Aware Feature Modulation for Cross-Domain Adaptation with Image-Only Inference

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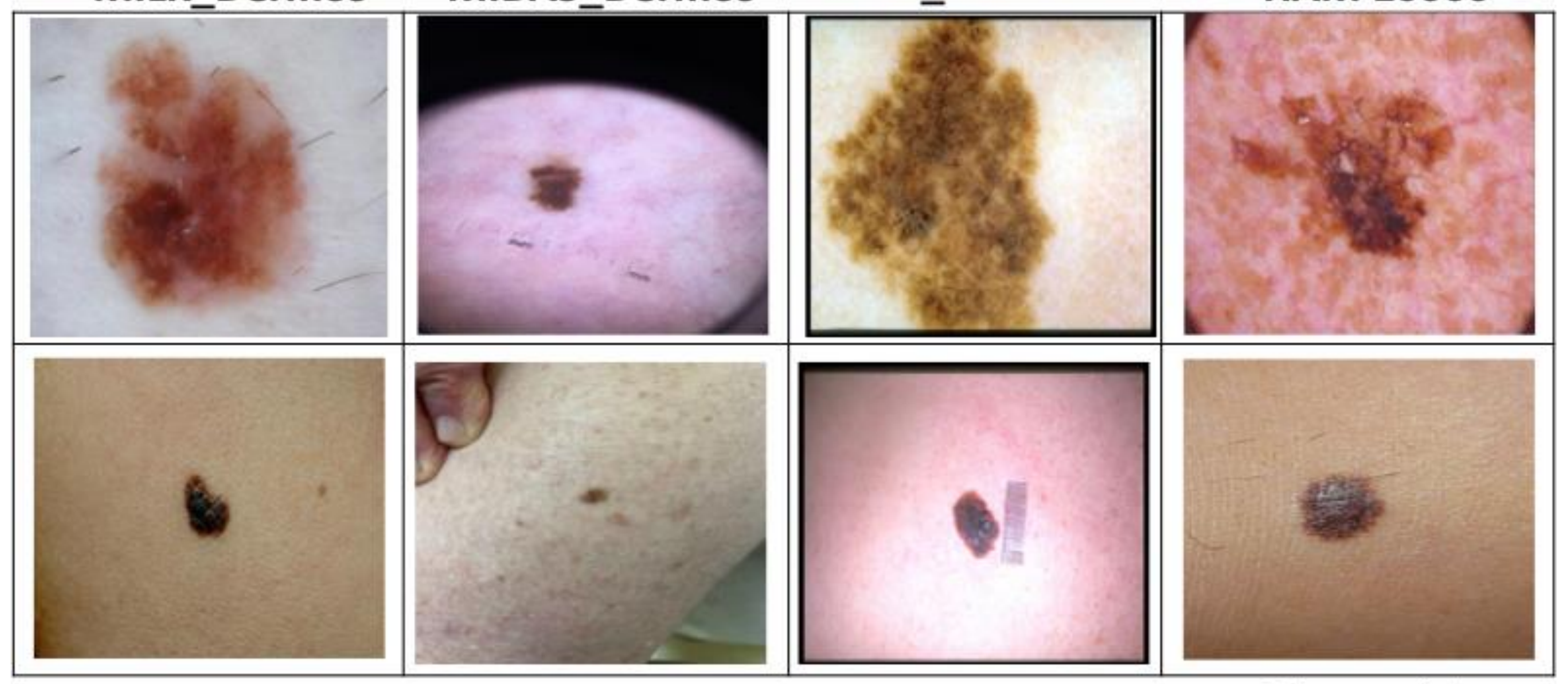
Problem Identification

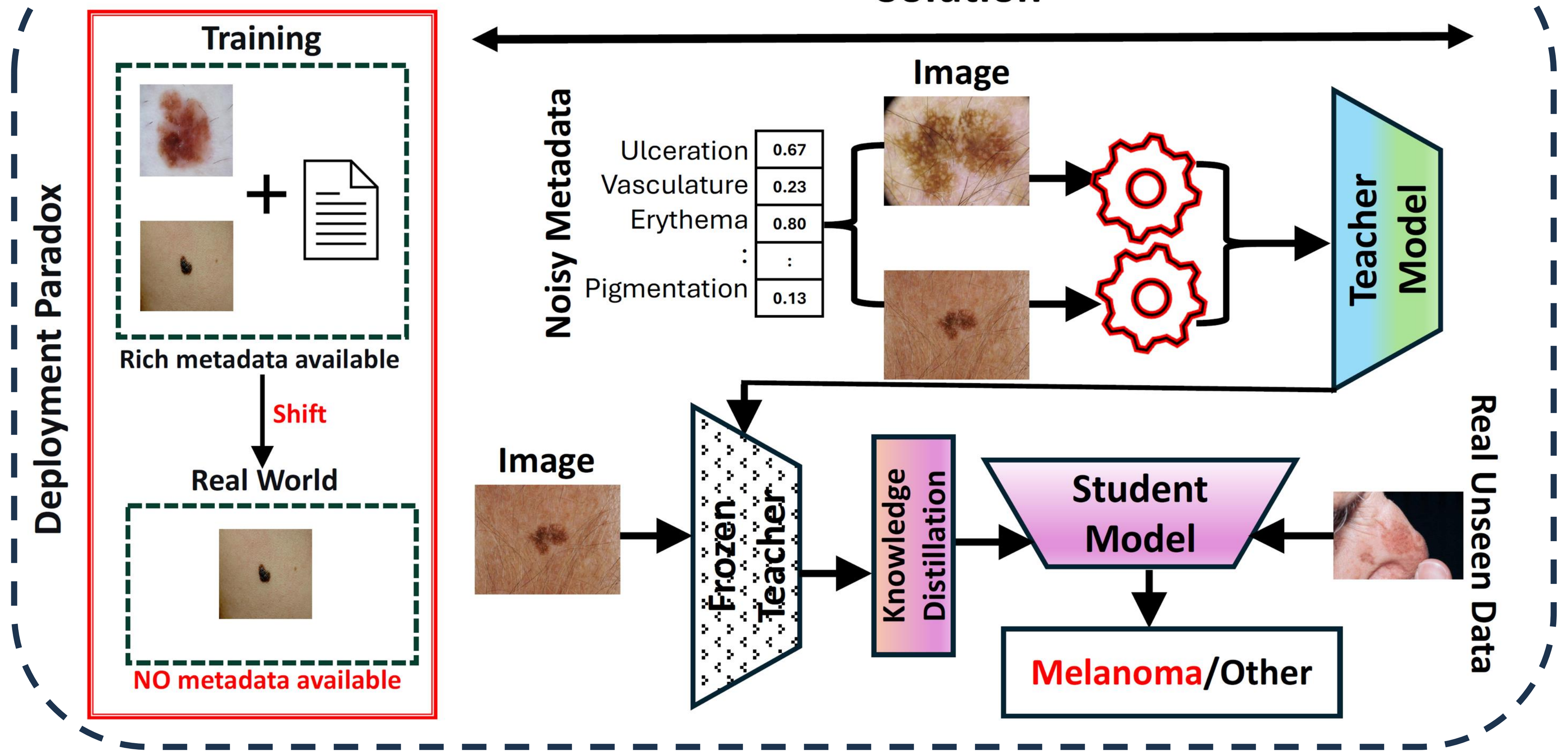
Dermoscopic → clinical domain shift is large.

Concepts help training.

Metadata is unavailable at deployment.

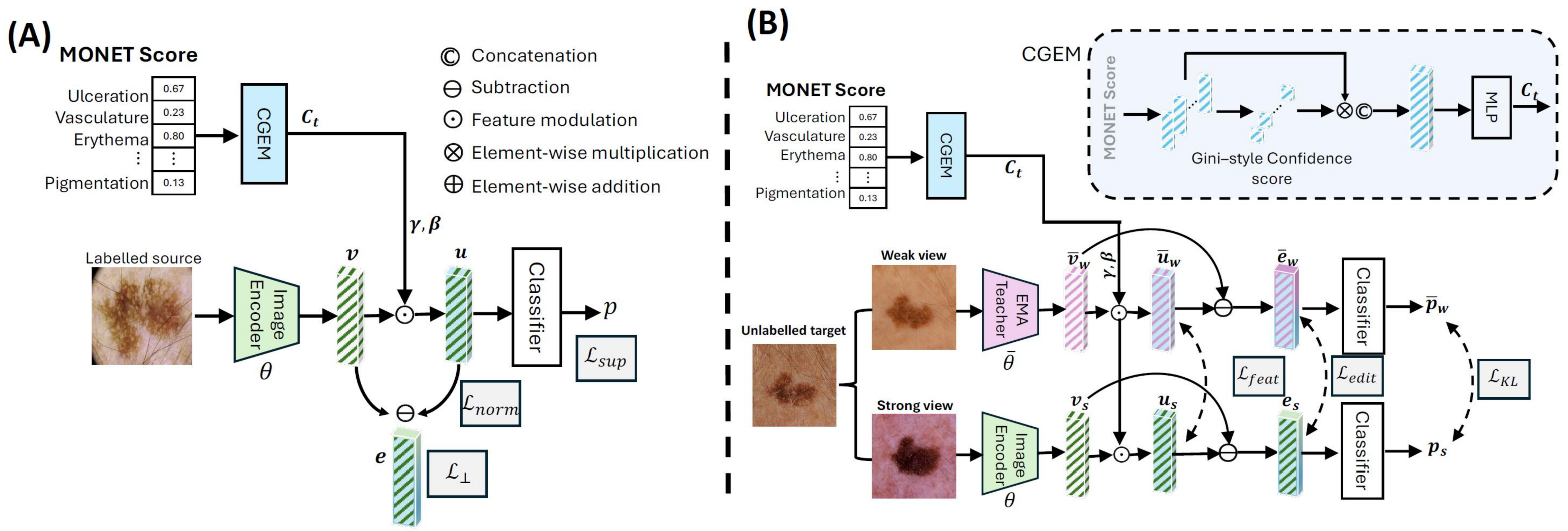
Dataset example



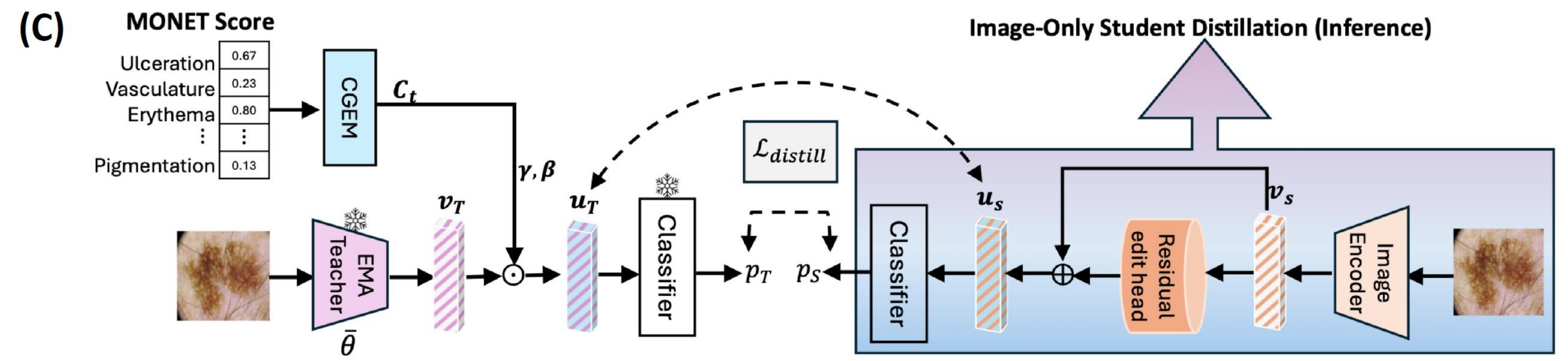


From Deployment Paradox to Image-Only Inference

Solution



Overall training objective $\mathcal{L} = \mathcal{L}_{sup} + \lambda_{\perp}\mathcal{L}_{\perp} + \lambda_{norm}\mathcal{L}_{norm} + \lambda_{kl}\mathcal{L}_{KL} + \lambda_{feat}\mathcal{L}_{feat} + \lambda_{edit}\mathcal{L}_{edit}$



CoFiDA-M in 3 Steps

A. Teacher: Image + MONET concepts → concept-edited features

B. Student: Distils teacher logits/features → image-only model

C. Deployment: Image only, no metadata → melanoma / other

Contacts & Resources

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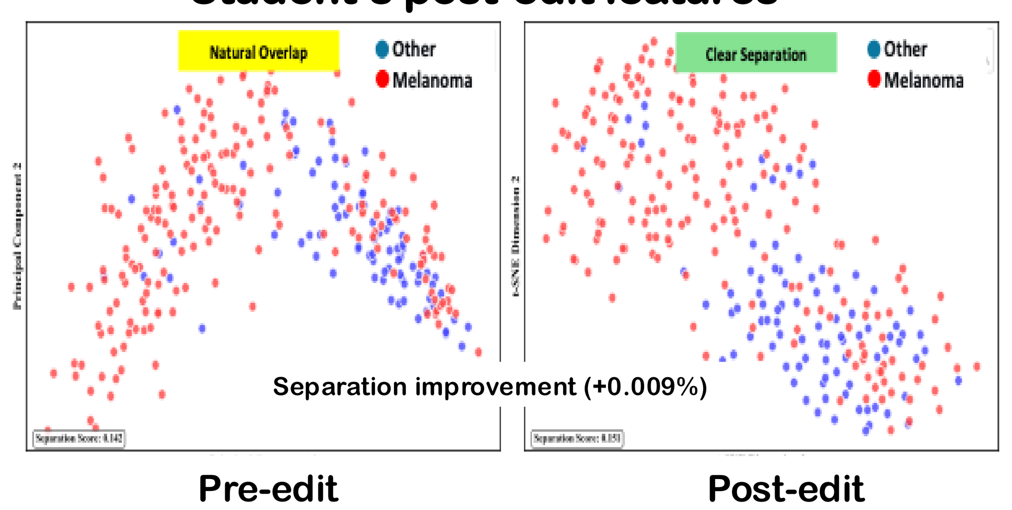
Code Repository — Scan for code

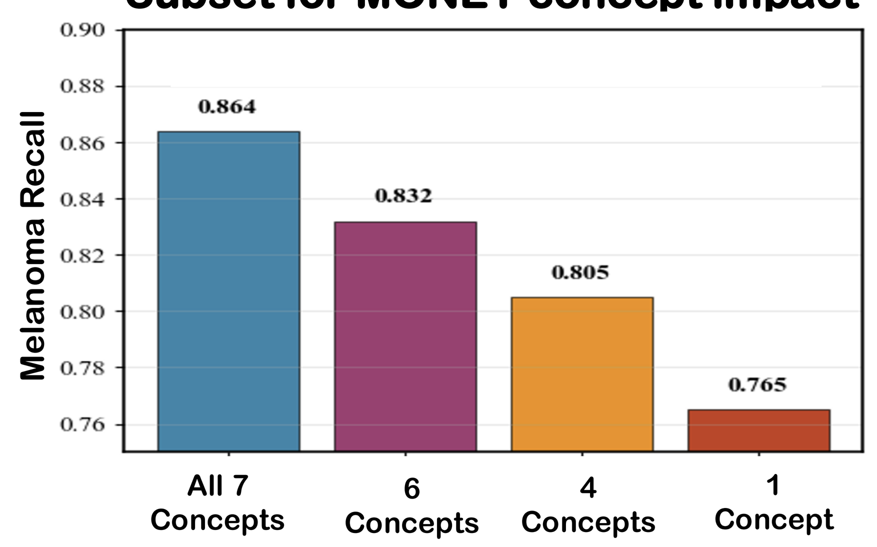
Other Project — Scan for details

Results

Setting / Method	AUROC Derm.	AUROC Clin.	Melanoma Recall Derm.	Melanoma Recall Clin.	Balanced Accuracy Derm.	Balanced Accuracy Clin.
Classical / Standard						
DANN	73.04	58.08	53.09	23.84	59.26	50.25
DeepCoral	69.50	56.46	69.50	26.91	61.99	54.05
MMD	74.08	58.67	59.23	18.31	67.60	53.27
MeanTeacher	55.97	48.91	64.02	77.67	46.33	47.93
IT-RUDA	75.85	56.52	30.62	12.92	68.47	52.69
Modern Source Free						
SHOT++	60.35	50.84	61.17	57.58	60.33	51.21
DINE	71.83	54.41	14.96	2.29	71.51	62.92
SFDA	56.08	47.25	21.62	10.56	49.82	50.86
CPD	61.51	48.23	56.63	60.74	51.66	50.52
Modern Test Time						
CoTTA	56.61	52.25	64.05	58.11	53.14	48.22
TENT	69.70	62.32	65.74	55.65	66.84	56.95
WoC	60.06	48.98	48.41	28.33	63.39	52.12
Multimodal						
DAMP	75.50	55.48	24.90	5.83	62.66	50.95
DALUPI	76.07	54.54	40.11	9.72	66.39	51.25
Source Only	69.96	58.39	63.55	55.77	61.00	54.00
Ours (CoFiDA-M)	**76.50**	**67.50**	**84.92**	**77.89**	**73.06**	**68.87**

Average unseen performance (%)





- CoFiDA-M gives the strongest image-only results on unseen datasets.
- Post edit features show clearer melanoma and other separation.
- Melanoma recall is highest when all 7 concepts are used.